

Supplemental Material

Control of nano-scale friction on gold in ionic liquids through applied potentials

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This set of Supplemental Material presents force vs. distance profiles recorded in [Py_{1,4}]FAP on Au(111), both using sharp AFM tips and colloid probes attached to Atomic Force Microscopy (AFM) cantilevers. Detailed analysis of the profiles allows to determine the nature of the interface layer that remains between probe and surface at all loads applied in this study.

Figure 1 presents force (F_{norm}) versus distance profiles for [Py_{1,4}]FAP confined between an approaching silica colloid probe (5 μm diameter) and the Au(111) electrode surface for different applied potentials. Force profiles can be used to elucidate the likely structure of ILs confined between solid surfaces, including the ion species in contact with the substrate [1, 2].

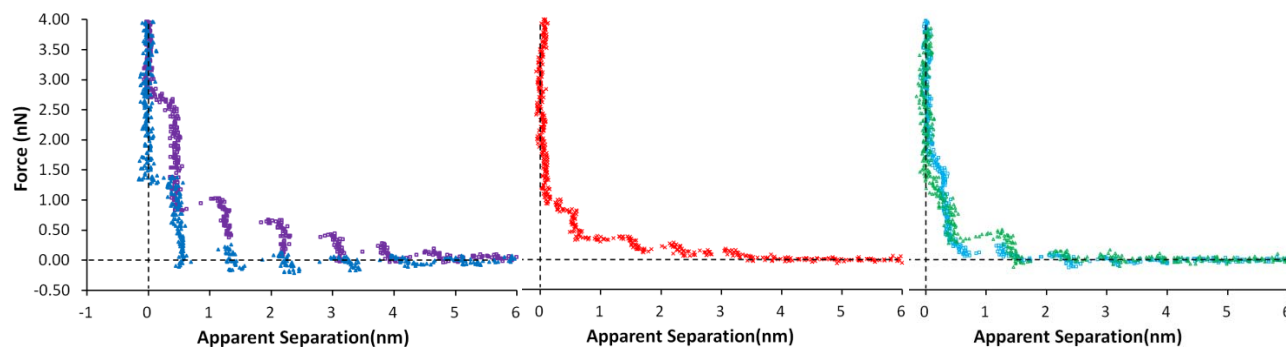


Fig. 1. Force versus distance profiles for an 5 μm colloid probe AFM tip approaching Au(111) surface in $[\text{Py}_{1,4}]\text{FAP}$ at -2 V (purple), -1 V (blue), Open Circuit Potential measured as 0.16 V (red), +1 V (green) and +1 V (light blue). All voltages are measured versus a Pt electrode, which is sufficiently stable under the conditions of these experiments.

The AFM force profiles in Figure 1 are qualitatively similar at all potentials, with stepwise force-distance data obtained. Assuming lamellar-like ordering, the width of each step must be determined by the physical dimension of one of the following: the cation, the anion, the ion pair, or combinations thereof. Step widths thus provide an indication of the composition of the layer. Higher force is required to rupture the steps at smaller tip-surface separations because the level of order in ion layers increases closer to the surface [1-5]. Many ionic liquids with C_4 alkyl chains have micellar [6] or sponge like [7, 8] nanostructure in the bulk. Analogously to surface induced sponge to lamellar phase transitions that occur in aqueous surfactant systems [9], the curved IL nanostructure is transformed to a flatter, more ordered, lamellar-like arrangement by the presence of the smooth solid surface, and the rupture force required to displace the layer increases.

In this context the force profiles in Figure 1 can be explained. For the OCP data, zero, constant force is recorded beyond ~ 5 nm as the AFM probe is not deflected by the approach of the

opposite surface, indicating that the AFM measurement is insensitive to any nanostructures that might exist in the [Py_{1,4}]FAP bulk. The probe encounters probe the first detectable layer at a separation of ~4.2 nm and pushes against it. At 0.07 nN force, this layer is ruptured, and then the tip jumps 0.9 nm before encountering another layer at ~3.3 nm from the interface. The process is repeated a further three times with layers detected at 2.4, 1.5 nm and 0.6 nm; this step spacing is in excellent agreement with the predicted [Py_{1,4}]FAP ion pair diameter of 0.9 nm determined from the bulk density [2]. The magnitude of the push-through forces for each ion pair layer increases as the separation decreases, confirming that the IL structuring becomes more pronounced with increasing confinement. This is consistent with other experiments [1-4, 10] and theoretical predictions [5]. When a bias (either positive or negative) is applied to the gold surface the general behaviour is similar to the OCP data, but the larger number of 0.9 nm steps, and the increased step height indicate that the degree of ordering of the confined film is significantly enhanced. It is worth noting that the zero of separation in such AFM colloid probe measurements is arbitrary, and determined by the position of closest approach.[11] Often this corresponds to physical contact of the surfaces, but can also correspond to the contact of bound layers when the maximum force exerted is less than the final push through force. [12, 13]. As will become apparent below, there must be a bound layer of ions still attached to the gold surface at the closest separation reached in our experiments.

The composition of the layer in contact with the gold surface is of greatest importance for the frictional response. It can be determined indirectly from the width of the step nearest the substrate in the AFM force profiles. Innermost steps of ~0.35 nm are detected at positive potentials (+1 V and +1.5V), compared to ~0.5 nm at negative potentials (-1 V and -2 V) and 0.6 nm at open circuit potential (ocp). All three distances are smaller than the dimension of the ion

pair. The size of the steps are consistent with layers that are enriched in *cations* at +1 V and +1.5 V and enriched in *anions* for -1 V, -2 V and at ocp. At first glance, this appears counterintuitive, as it suggests that ions of the same charge (co-ions) as the surface are in contact with it. This strongly implies that, as flagged above, the AFM colloid probe and the gold surface are *not* in contact at smallest separation and that indeed the closest approach in these experiments occurs when the colloid probe contacts an ion layer that is unable to rupture. (In these experiments normal forces were measured up to applied loads of 40 nN). The bound layer is enriched in ions of opposite charge to the gold substrate. The ionic nature of the liquid leads to the screening of long range electrostatic forces. Thus, the effect of the applied potential is to polarise the ion pairs closest to the surface such that counter-ions are enriched (attracted) and co-ions are depleted (repelled). The effect of this polarisation is to effectively split the adsorbed ion pairs into two layers, which become more distinct as the magnitude of the applied potential increases, with correspondingly smaller cohesion. This leads to the possibility of the co-ion layer being displaced at high forces, but not the closest, counter-ion layer. It is thus the expected thickness of the counter-ion layer which is detected as the height of the last step upon approach.

Force versus distance profiles for the sharp AFM tip are shown in Fig. 2. For a potential of -1.5 V, the result and analysis are exactly the same as discussed above for the colloid probe at negative potentials. There are several steps in the approach of 0.9 nm height, indicating the squeeze-out of ion pair layers. The last step upon approach has a height of only 0.6 nm, indicating that only the anions are squeezed out and that the closest approach is towards a bound cation layer. In contrast, at a potential of -1.3 V the tip approach exhibits a two steps, whose heights sum up to 0.9 nm. At this potential, both anion and cation layer are penetrated by the sharp AFM tip and direct contact with the Au(111) surface is established.

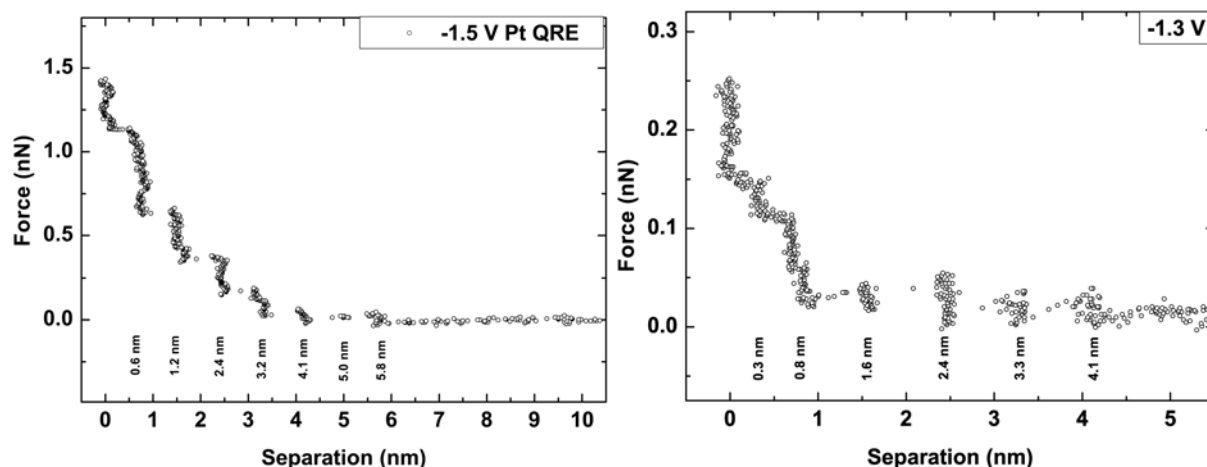


Fig. 2. Force versus distance profiles for a sharp AFM tip approaching Au(111) surface in [Py_{1,4}]FAP at a potential of -1.5 V and -1.3 V measured versus a Pt quasi reference electrode.

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